



Calculations/Steam Plant
 Project: **HVAC System Upgrade & Exp. B44**
 VA Project Number: 546-22-110
 Above Group Project Number: 0122008
 Current Phase: **Temporary Boiler Sizing**

PLANT CALCULATIONS FOR:
MIAMI VA MEDICAL CENTER
HVAC SYSTEM UPGRADE & EXP. B44

April 29, 2026

Prepared By: Geoff Chenoweth

Reviewed By: Geoff Chenoweth

Approved By: Geoff Chenoweth

No.	Revision	Date
1	Requested Change by VA	4/29/2026

HEADQUARTERS
 305 East Drive Suite H
 Melbourne, FL 32904
 (321) 345-9026

111 North Orange Ave.
 Suite 800
 Orlando, FL 32801
 (407) 377-0200

70 Birch Alley
 Suite 240
 Beavercreek, OH 45440
 (937) 795-4440

170 Commerce Way
 Suite 200
 Portsmouth, NH 03801
 (603) 261-2064

400 East Pratt Street
 8th Floor
 Baltimore, MD 21202
 (443) 788-4200



Calculations/Steam Plant
 Project: **HVAC System Upgrade & Exp. B44**
 VA Project Number: 546-22-110
 Above Group Project Number: 0122008
 Current Phase: **Temporary Boiler Sizing**

PROJECT SUMMARY

SYSTEM FLOWS:

		PPH	GPM	Cond. Return Rate	Notes:
Softwater Required	SWr	22,984	46	0%	Maximum Softwater Demand
Returned Condensate	COND	25,213	52	100%	Maximum Condensate Return
Steam to Deaerator	STEAM	4,674	---	0%	Low Pressure Steam To The Deaerator
Steam to Campus	StC	25,213	---	100%	Maximum Steam to Campus
Feedwater	FEED	27,658	58	0%	Deaerator Discharge

	Recirc %	Total Flow GPM	Head TDH	Motor Power HP
Condensate Pump		30	130	2
90 Feedwater Pump		40	480	10
250 Feedwater Pump		80	820	25

Feedwater Temp	227	°F
Enthalpy of Water	195	BTU/LB
Steam Operating Pressure	100	PSIG
Enthalpy of Steam	1190	BTU/LB
Boiler Capacity	33475	BTUH/HP
Boiler Rating	500	HP
Nom. Boiler Cap. @ 212°F	16738	PPH
Actual Boiler Cap. @ Pressure	16830	PPH
Qty. of Boilers Running	1	
Total Capacity	16830	PPH

**80 % CONDENSATE RETURN
170° F CONDENSATE
Design Condition for Condensate System**

INPUT DATA:

DATA	SYMBOLS	VALUE	UNITS	SOURCE	NOTES
Boiler Steam Output	SO	16,830	PPH	GIVEN	
Continuous Blowdown Rate	CBD	3%	%	ASSUMED	
Campus Condensate Return Rate	CCrr	82%	%	GIVEN	
Average Condensate Temperature	CDt	170	°F	GIVEN	
Average Softwater Temperature	SWt	70	°F	GIVEN	
Deaerator Pressure	Dap	5	PSIG	KNOWN	
Parasitic Steam Load	PAR	2,000	PPH	ASSUMED	
Desuperheating Flow	DeSup	0	PPH	ASSUMED	

EQUATIONS:

Steam to Campus	StC	SO - STEAM - PAR
Feedwater Flow	FEED	DeSup + SO + (SO* CBD)
Returned Condensate	COND	StC * CCrr

Conservation of Mass Equation: $\dot{m}_{COND} + \dot{m}_{SOFT} + \dot{m}_{STEAM} = \dot{m}_{FEED}$

Conservation of Energy Equation: $\dot{m}h_{COND} + \dot{m}h_{SOFT} + \dot{m}h_{STEAM} = \dot{m}h_{FEED}$

CALCULATIONS:

		MASS	TEMP	ENTHALPY	ENERGY
	SYMBOL	<i>m</i>	<i>T</i>	<i>h</i>	<i>mh</i>
CONDENSATE	COND	12,733	170	138	1,756,965
SOFTWATER	SOFT	3,300	70	38	125,652
PUMPED CONDENSATE	MU	16,032	149	117	1,882,618
STEAM	STEAM	1,302	227	1,156	1,505,164
FEEDWATER	FEED	17,335	227	195	3,387,782

RESULTS:

Steam to Campus	StC	15,528	PPH	CALCULATED
Softwater Required	SOFT	3,300	PPH	6.6
Returned Condensate	COND	12,733	GPM	26.1
Returned to the Deaerator	MU	16,032		32.7
Feedwater	FEED	17,335		36.4

* - Calculated on different sheet.

Assumptions:

- Deaerator vent losses not accounted for.
- Deaerator Steam is assumed to be saturated.

0 % CONDENSATE RETURN
170° F CONDENSATE
Design Condition for Softwater System

INPUT DATA:

DATA	SYMBOLS	VALUE	UNITS	SOURCE	NOTES
Boiler Steam Output	SO	16,830	PPH	GIVEN	
Continuous Blowdown Rate	CBD	3%	%	ASSUMED	
Campus Condensate Return Rate	CCrr	1%	%	GIVEN	
Average Condensate Temperature	CDt	170	°F	GIVEN	
Average Softwater Temperature	SWt	70	°F	GIVEN	
Deaerator Pressure	Dap	5	PSIG	KNOWN	
Parasitic Steam Load	PAR	0	PPH	ASSUMED	
Desuperheating Flow	DeSup	0	PPH	KNOWN*	

EQUATIONS:

Steam to Campus	StC	SO - STEAM - PAR
Feedwater Flow	FEED	DeSup + SO + (SO*CBD)
Returned Condensate	COND	StC * CCrr

Conservation of Mass Equation: $\dot{m}_{COND} + \dot{m}_{SOFT} + \dot{m}_{STEAM} = \dot{m}_{FEED}$

Conservation of Energy Equation: $\dot{m}h_{COND} + \dot{m}h_{SOFT} + \dot{m}h_{STEAM} = \dot{m}h_{FEED}$

CALCULATIONS:

		MASS	TEMP	ENTHALPY	ENERGY
	SYMBOL	<i>m</i>	<i>T</i>	<i>h</i>	<i>mh</i>
CONDENSATE	COND	139	170	138	19,205
SOFTWATER	SOFT	14,284	70	38	2,665
PUMPED CONDENSATE	MU	14,423	71	39	21,871
STEAM	STEAM	2,912	227	1,156	3,365,912
FEEDWATER	FEED	17,335	227	195	3,387,782

0

RESULTS:

Steam to Campus	StC	13,918	PPH	CALCULATED
Softwater Required	SOFT	14,284	PPH	CALCULATED
Returned Condensate	COND	139	GPM	CALCULATED
Returned to the Deaerator	MU	14,423	GPM	CALCULATED
Feedwater	FEED	17,335	GPM	CALCULATED

* - Calculated on different sheet.

Assumptions:

Deaerator vent losses not accounted for.

Deaerator Steam is assumed to be saturated.

